The Garden Watering Dilemma

PRESENTED BY

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INTRODUCTION

- ✓ Efficient water distribution is important for gardening.
- ✓ Manual watering causes over-/under-watering.
- ✓ Our project simulates a smart irrigation system using C, threads, semaphores & mutexes.
- ✓ Time runs from **6:00 AM to 10:00 PM** with dynamic conditions.

PROJECT GOALS

Create a **realistic irrigation simulation**.

Manage **multiple garden sections** independently.

Respect **daily water limits**.

Use **priority-based watering**.

Implement **soil moisture logic**.

Generate a **final report of performance**.

WHY MULTITHREADING?

- Each garden section behaves like an independent process.
- Concurrency helps simulate real-time irrigation.
- Threads run in parallel:
- Item Threads → watering logic
- Time Thread → manages the day clock
- Mutexes prevent data conflicts.
- Semaphores control scheduling.

THREAD ARCHITECTURE

Time Thread

- Simulates time from 6 AM → 10 PM
- Releases semaphores periodically
- Dries soil randomly
- Prints hourly water usage
- Stops everything at 22:00

Item Threads

- One thread per garden section
- Executes watering logic when signaled
- Checks moisture, priority, water availability
- Maintains 1-hour minimum gap
- Updates soil state

SYSTEM COMPONENTS

Each garden section includes:

- Name
- Required waterings
- Completed count
- Priority (High/Med/Low)
- Soil moisture state
- Last watering time
- Thread + semaphore
- Per-section mutex

Global components include:

- Time simulation
- Water limit
- Shared mutexes for safety

WATERING LOGIC

A section is allowed to water only if:

1. Soil is dry
2. At least 1 hour passed since last watering
3. Water limit is not exceeded
4. High priority sections get preference
5. Current time is before 10 PM
6. Moisture checks allowed only before 8 PM

SAMPLE OUTPUT SCREENSHOT

```
sabik@sabik-VirtualBox:~$ cd Desktop
sabik@sabik-VirtualBox:~/Desktop$ cd Labtest01
sabik@sabik-VirtualBox:~/Desktop/Labtest01$ gcc GW.c -o GW
sabik@sabik-VirtualBox:~/Desktop/Labtest01$ ./GW

WELCOME

1. Start Garden Simulation
2. Exit
Choose an option:
```

```
1. Start Garden Simulation
2. Exit
Choose an option: 1
Enter number of garden sections: 5

Section 1:
  Name: Flower
  Daily waterings needed: 3

Section 2:
  Name: Fruits
  Daily waterings needed: 4

Section 3:
  Name: Vegetable
  Daily waterings needed: 3

Section 4:
  Name: Herbs
  Daily waterings needed: 2
```

```
FINAL WATERING REPORT
========================================
Section         Required   Done       Priority   Status
------------------------------------------------------------
flower          3          3          MEDIUM     COMPLETE
fruits          4          3          LOW        PARTIAL
vegetable       3          2          LOW        PARTIAL
herbs           2          2          LOW        COMPLETE
greenhouse      2          2          HIGH       COMPLETE
------------------------------------------------------------
TOTAL:          14         12
Water used:     12/12 sessions
Efficiency:     85.7%

FAILURE: Significant unmet requirements

ANALYSIS:
- Used all available water
- Failed to meet 2 watering requirements due to:
  1. Time constraints
  2. Moisture sensor blocks
  3. Priority system
  4. 1-hour gap requirement

=== Simulation complete ===
```

RESULTS

- **Final report includes:**
  - Required vs completed waterings
  - Priority levels
  - Efficiency %
  - Water usage summary
  - Sections marked: COMPLETE, PARTIAL, FAILED

- **System shows realistic limits:**
  - Soil too wet → skip watering
  - Water finishing early
  - Priority handling
  - Time restriction

CONCLUSION

- Successfully implemented a realistic irrigation simulator.
- Demonstrated multithreading, mutex locks & semaphores.
- Highlighted environmental constraints & resource management.
- System behaves like a real smart irrigation controller.
- Strengthens understanding of OS concepts like scheduling & synchronization.

FUTURE IMPROVEMENTS

- Real soil sensors (IoT)
- Weather-based watering decision
- Graphical dashboard
- Machine-learning based moisture prediction
- Better scheduling algorithm (EDF / Round Robin / Priority Queue)

THANK YOU!